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An AI-Driven Web-Based Interactive Sign Language Learning Platform for Deaf and Hard-of-Hearing Children

A Project Proposal by

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ABSTRACT

Deaf and hard-of-hearing (DHH) children face significant barriers to acquiring literacy and communication skills due to the lack of interactive sign language learning tools. Existing systems are largely limited to static content or alphabet recognition with little emphasis on real-time corrective feedback, word building through fingerspelling, or the use of domain-specific vocabulary signs. This shortage leaves students with fragmented skills and no means to transition from letter learning to effective everyday communication.

In order to address this challenge, the project develops an interactive web-based learning platform using deep learning and computer vision for real-time sign detection. Using MediaPipe for hand landmark detection and CNNs for classification, the system recognizes manual alphabet signs and supports progressive learning stages like alphabet practice, word building through fingerspelling, introduction of dedicated object-based signs, and mixed practice for reinforcement. The platform provides personalized corrective feedback, gamified exercises, and progress tracking to enable engagement, retention, and motivation. Evaluation will be based on recognition accuracy, latency, robustness with different users and settings, and learning outcomes via pre- and post-learning testing.

Subject Descriptors:

● Human-centered computing → Accessibility technologies → Accessibility systems and tools

● Computing methodologies → Computer vision → Vision for recognition → Gesture recognition

● Applied computing → Interactive learning environments → Computer-based instruction

Keywords: Sign language learning, gesture recognition, deep learning, computer vision, corrective feedback, inclusive education, web-based platform

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Table of Abbreviations

CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
CTC	Connectionist Temporal Classification
TCN	Temporal Convolutional Network
DHH	Deaf and Hard of Hearing
ASL	American Sign Language
ISL	International Sign Language
IU/UX	User Interface / User Experience

CHAPTER 01: INTRODUCTION

1.1 Chapter Overview

The background, problem definition, and research motivation are presented in this chapter, which kicks off the research project. The chapter outlines the purpose and goals of the study, identifies the research gap, and lists anticipated contributions. Principal challenges are also covered in this chapter. Mapping objectives to learning outcomes and research questions brings the chapter to a close.

1.2 Problem Background

A key component of social inclusion and a fundamental human right is education. However, children who are deaf or hard of hearing (DHH) still face barriers to accessing high-quality education. Over 34 million children worldwide suffer from debilitating hearing loss, and more than 80 percent of them do not have access to sufficient educational resources, according to the World Health Organization (BeMyVoice, 2024). Due in large part to a lack of qualified teachers, a lack of institutional support, and overcrowded special education facilities, research indicates that over half of children with communication disorders never attend school in Sri Lanka (Dodandeniya et al., 2023).

The absence of a nationally accepted sign language and limited availability of interpreters are two primary barriers that prevent DHH children from having access to schooling (Daily Mirror, 2015; BeMyVoice, 2024). Due to inadequate teaching material and trained teachers, individuals who are deaf take rarely in attaining literacy and communication skills, which are needed for social integration and academic progress (Deoghare et al., 2024).

Even though DHH learners primarily communicate through sign language, the pedagogical tools that are now available, such dictionaries, flashcards, and movies, are passive and do not support active learning. Students are left with gestures that they memorise without knowing the context in which they are employed since these resources just display vocabulary without offering real-time feedback or an interactive experience (Fernando et al., 2025; Milinda et al., 2024). Additionally, class-based education depends on a lot of teacher supervision, which is not feasible when there are

only a few qualified experts available (Jayawardana et al., 2023). As a result, students are not given the chance to practise independently and get little assistance in the form of corrective feedback.

Recent advancements in artificial intelligence and computer vision offer promising solutions to address these limitations. Convolutional Neural Networks (CNNs) combined with hand-tracking frameworks such as MediaPipe can accurately detect and classify sign gestures in real time (Sau et al., 2022; Jayawardana et al., 2023). Research has demonstrated that deep-learning-based gesture recognition models can achieve high accuracy even under varied conditions, enabling adaptive and responsive learning environments (Milinda et al., 2024; Fernando et al., 2025). When integrated into a web-based interactive learning platform, these technologies allow learners to receive instant feedback, practice autonomously, and engage in immersive learning sessions (Deoghare et al., 2024; Dodandeniya et al., 2023).

Therefore, the context of this problem highlights the urgent need for AI-driven, web-based, interactive sign language learning platforms that transcend static pedagogical methods. Such systems can empower DHH children to develop literacy and communication skills more effectively, promoting inclusion, independence, and equitable educational access (Fernando et al., 2025; BeMyVoice, 2024).

1.3 Problem Definition

This study's primary focus is the dearth of interactive online resources that provide real-time recognition and feedback modification for sign language instruction. Nowadays, most online resources are used as reference materials or as static resources, which leaves little opportunity for students to study, practice, and improve their abilities in real time. As a result, even when kids are taught signs, they won't be able to use them in word-building exercises or duplicate them correctly.

Additionally, the majority of current systems are limited in their scope and just focus on alphabet recognition; they are unable to provide scaffolded learning, which allows kids to go from individual letters to entire words. This maintains students in a situation where they lack the reinforcement necessary to effectively recall and apply their knowledge, resulting in an exposure-to-mastery gap.

Thus, an online program that combines live gesture detection, practice, and feedback with sign language instruction is desperately needed. In addition to teaching signs, it would let kids practice actively, get feedback when they make mistakes, and learn to read interactively and step-by-step.

1.3.1 Problem Statement

There is a lack of an interactive AI-driven web-based platform that enables deaf and mute children to learn sign language through real-time gesture recognition and personalized feedback.

1.4 Research Motivation

Millions of children worldwide struggle to communicate because of hearing problems, yet there aren't many educational resources that are tailored to meet their unique learning requirements. Traditional teaching resources frequently rely on the teacher, which restricts their scalability and accessibility. Learning sign language can become an interesting, self-directed experience with the help of an AI web platform that can recognize gestures in real time and provide interactive feedback. The method can boost confidence, increase literacy, and promote involvement in social and educational environments by letting students practice on their own with prompt correction (Sau et al., 2022).

1.5 Existing Work

Citation	Summary	Limitation	Contribution
Dodandeniya et al. (2023) – Visual Kids	Built interactive learning app with gamification for deaf children.	Focused mainly on alphabets and numbers.	Proved gamified tools improve engagement.
Daily Mirror (2018) – Sri Lanka Terribly Short of Sign Language Interpreters	Reports on the severe shortage of professional sign language interpreters in Sri Lanka, leaving many deaf students without access to proper education.	Journalistic piece, not a technical or system-focused study.	Highlights the urgent demand for technology-based alternatives to fill interpreter gaps in Sri Lanka.
A Review of Real-Time Sign Language Recognition for Virtual Interaction on Meeting Platforms (2024)	Reviews CNN, RNN, multimodal fusion, depth sensors, and mobile approaches for real-time SLR in online meetings (Zoom/Teams). Highlights challenges such as camera angle, lighting, and dataset shortages.	Focused on meeting platforms, not educational contexts; lacks experimental implementation for children's learning.	Provides a comprehensive overview of best-performing models (CNNs, RNNs, multimodal) and their applicability to real-world virtual environments.
Sign Language Detection in	Proposed a CNN-based real-time system using TensorFlow and	Small dataset, sensitive to lighting/backgrounds,	Demonstrated low-cost, webcam-based real-time recognition with

Citation	Summary	Limitation	Contribution
Real-Time Applications (Milinda et al., 2024, IEEE ICCIRT)	OpenCV to detect ISL gestures via webcam, achieving ~87.6% accuracy	focused only on basic gestures.	high accuracy and accessibility.
Deoghare et al. (2024) – Interactive Platform for Literacy	Web-based platform using MediaPipe and CNNs for alphabets, numbers, and emotions.	Accuracy ~88%, struggles with similar letters.	Demonstrated feasibility of web platforms with real-time feedback.

Table 1: Existing Work

1.6 Research Gap

Despite recent advancements in sign language recognition and learning technologies, most existing applications continue to rely on static teaching methods such as dictionaries, flashcards, and video tutorials. While these resources help learners gain initial exposure to sign vocabulary, they fail to promote active engagement or provide real-time corrective feedback (Deoghare et al., 2024; Dodandeniya et al., 2023). As a result, children often learn signs through passive observation rather than interactive practice, leading to limited mastery and difficulty applying signs in meaningful communication.

Although some studies have integrated gesture recognition into educational tools, these systems are often restricted to basic alphabet detection and lack structured learning pathways that help users progress from individual letters to complete words or phrases (Milinda et al., 2024; Fernando et al., 2025). This limitation prevents learners from developing both literacy and fluency in sign communication. Furthermore, many models perform well in laboratory environments but fail to maintain accuracy in real-world conditions, where differences in lighting, background, and hand shape introduce high variability (Jayawardana et al., 2023; Sau et al., 2022).

A notable gap also exists in the development of web-based interactive learning platforms that leverage modern deep learning and computer vision techniques for sign language education. While several mobile and desktop applications have shown potential, most lack scalability, browser compatibility, and inclusive design suited for children (Jayasuriya et al., 2020; BeMyVoice, 2024). Few systems have addressed the crucial educational transition from fingerspelling words to using single, meaningful signs that represent everyday objects and actions (Deoghare et al., 2024; Fernando et al., 2025).

Therefore, the research gap lies in the absence of an interactive, web-based sign language learning platform that integrates real-time gesture recognition, personalized corrective feedback, and a structured progression from alphabet learning to vocabulary-based communication. Bridging this gap will transform sign language education from passive exposure into active, self-directed learning, empowering deaf and hard-of-hearing children to achieve both literacy and natural communicative competence (Dodandeniya et al., 2023; Jayawardana et al., 2023).

1.7 Contribution to Body of Knowledge

Problem Domain Contribution:

The proposed system addresses barriers in sign language education by providing a web-based interactive learning platform where children can practice gestures and receive real-time corrective feedback. Unlike static tools, the platform enables scaffolded learning: beginning with alphabets, moving to word spelling, and finally introducing dedicated vocabulary signs, thereby fostering literacy, communication, and confidence.

Research Domain Contribution:

From a research perspective, this project contributes to the field of deep learning–based sign recognition by applying lightweight convolutional neural networks (CNNs) integrated with real-time hand tracking (e.g., MediaPipe). By focusing on a web-based deployment, it demonstrates how sign recognition models can be adapted for scalability, accessibility, and inclusiveness. The work advances existing research by shifting from isolated recognition experiments toward applied

interactive learning environments that support both accuracy and usability in real-world educational contexts.

1.8 Research Challenges

The creation of an interactive web-based sign language learning platform has a number of technical and pedagogical issues that need to be considered in order to make it effective and inclusive at the same time. The main problem is to balance recognition accuracy and performance in real time of the system. CNNs and LSTMs are computationally expensive deep learning models, and to achieve low latency with precision, it is necessary to employ advanced optimization methods and model compression designs (Sau et al., 2022; Milinda et al., 2024). Moreover, the variability of gestures used by children due to the variability in the size of the hands, orientation, and the speed of the signing contributes its part to the difficulty of recognition. The system should be able to generalize on varying user inputs and yet have a dependable accuracy on prediction (Jayawardana et al., 2023; Fernando et al., 2025).

Such environmental discrepancies as bad lighting, background noises, and inconsistent quality of cameras also have a negative impact on model performance, particularly when it is used in uncontrollable environments, such as classrooms or households. Most studies also show encouraging results at lab settings, but they tend to reduce in strength when applied to the real world (BeMyVoice, 2024; Dodandeniya et al., 2023). The other major impediment is the absence of large, high quality, and context-specific sign language databank, especially those that are regional like the Sri Lankan Sign Language. Such a shortage requires the use of data augmentation or transfer learning methods in order to enhance the generalization of the models (Deoghare et al., 2024; Fernando et al., 2025).

It is also necessary to make the user's experience interesting and user-friendly. To create a child-friendly interface, it is necessary to implement gamified components, visual rewards, and adaptive feedback to maintain the attention and motivation of the learners during their practice (Dodandeniya et al., 2023; Deoghare et al., 2024). Technically, the device, browser, and internet conditions scalability should also be ensured without the use of dedicated hardware since the

modern high-performance models are frequently highly computational (Jayasuriya et al., 2020; Milinda et al., 2024). Additionally, the system is bound to handle two recognition problems, namely correctly processing both static fingerspelling and dynamic object-based gestures, which will involve more complex temporal modeling processing sequential data (Sau et al., 2022; Jayawardana et al., 2023). Lastly, the systematic development of a learning pathway, which helps children to transition through the knowledge of the alphabet and eventually to word building and contextual object signs, is a challenging pedagogical task. Such consistency between the AI-based recognition and the instructional design is very important so that comprehension, retention, and progressive skill acquisition take place (Deoghare et al., 2024; Fernando et al., 2025).

1.9 Research Questions

RQ1: How can a deep learning and computer vision-based approach be effectively used to implement real-time sign language gesture recognition for an accurate assessment and personalized corrective feedback in educational contexts?

RQ2: Which are the best multimodal interaction techniques (gesture recognition, object image capture, emotion recognition) that foster engagement, learning achievements, and communication abilities in deaf and mute children?

RQ3: What design and usability features are most effective in creating a child-friendly, web-based sign language learning platform that promotes active learning, motivation, and long-term retention?

1.10 Research Aim

The objective of this research is to create, develop, and assess a web-based interactive platform for learning sign language, utilizing deep learning and computer vision to provide real-time gesture recognition and tailored corrective feedback for children who are deaf or hard of hearing.

1.11 Research Objectives

	Research Objective	**Description**	**LOs Mapped**	**RQ Mapped**
Problem Identification	**R01: To identify the research gap in existing sign language learning systems**	Analyse limitations of current solutions (static resources, lack of feedback, limited scalability) to justify the need for a web-based interactive platform.	LO2, LO4	RQ1, RQ2, RQ3
Literature Review	**R02: To conduct a detailed review of deep learning and computer vision techniques for sign recognition**	Critically examine CNN, RNN, MediaPipe, and real-time SLR models to determine the most suitable approach.	L01, L04	RQ1
	R03: To review multimodal	Explore gesture recognition, object	L01, L04	RQ2

	Research Objective	Description	LOs Mapped	RQ Mapped
	interaction techniques for sign language learning	prompts, and engagement features that enhance learning outcomes		
	R04: To review design and usability principles for child-friendly educational platforms	Identify features and gamification strategies that promote motivation, retention, and inclusivity.	L01, L04	RQ3
Requirement Elicitation	**R05: To identify stakeholders and collect user requirements**	Engage with teachers, parents, and DHH learners to gather functional and non-functional requirements.	L03 , LO6	RQ1, RQ2, RQ3
System Design	**R06: To design a modular web-based architecture for the platform**	Define architecture integrating recognition, feedback, and learning modules with scalability in mind.	LO1, LO3	RQ1, RQ2
Implementation	**R07: To develop and train deep**	Build CNN/MediaPipe models for real-time	LO1, LO5, LO7	RQ1

	Research Objective	Description	LOs Mapped	RQ Mapped
	learning models for gesture recognition	alphabet and word recognition.		
	R08: To implement interactive practice modules with corrective feedback	Enable children to practice signs, spell words, and receive feedback to improve literacy and confidence.	L05, LO7	RQ1, RQ2
	R09: To integrate engaging UI/UX and gamification features	Add child-friendly design elements such as badges, progress tracking, and object-based learning.	LO5,LO7	RQ2, RQ3
Testing	**R10:** To evaluate the platform for accuracy, usability, and educational effectiveness	Test with datasets and users; measure recognition accuracy, latency, engagement, and learning gains.	LO1, LO2, LO6, LO7, LO8, LO9	RQ1, RQ2, RQ3
Documentation	**R11:** To produce comprehensive technical and research documentation	Document design, implementation, evaluation, and findings in a professional academic format.	LO2, LO4, LO6, LO8	-

	Research Objective	Description	LOs Mapped	RQ Mapped
Publication	**R12:** To publish findings in academic venues	Publish literature review and final results as research papers to contribute to the academic community.	LO4, LO6, LO8	-

Table 2: Research Objectives

1.12 Project Scope

1.12.1 In Scope

This research focuses on developing a web-based interactive platform for sign language learning. The system will primarily concentrate on:

- Teaching the manual alphabet (A–Z) using real-time hand gesture recognition.
- Allowing learners to practice word-building through fingerspelling, for example, spelling (C–A–T) by combining letters.
- Introducing learners to dedicated vocabulary signs for common objects, so they can move beyond letters to meaningful communication.
- Providing instant corrective feedback, helping children identify mistakes and adjust their gestures in real time.
- Incorporating gamification elements (points, badges, progress indicators) to keep learning enjoyable and motivating.
- Tracking individual progress so both learners and educators can monitor improvement over time.

- Evaluating system performance not only on technical accuracy and speed, but also on how engaging and effective it is for learners.

1.12.2 Out Scope

While the system aims to be as comprehensive as possible, a few areas remain outside the project's current scope due to time and resource constraints:

- The solution will be web-based only; developing a mobile application is not included at this stage.
- Features such as emotion recognition or hazard detection are excluded, as the focus is on learning sign language.
- The platform will initially support English/ASL-based signs; incorporating Sinhala Sign Language or other regional sign languages is left for future expansion.
- Integration with specialized assistive devices (e.g., hearing aids, AR/VR tools) is not considered in this version.
- Large-scale real classroom deployment will not be carried out; instead, testing will be limited to smaller pilot studies.

1.13 Hardware/Software Requirements

To build and evaluate the system effectively, both hardware and software resources are required. The specifications are chosen to balance accessibility for development with the higher computational needs of deep learning models.

Hardware Requirements:

Category	**Specification**
Development Machine (Minimum)	Intel Core i5 / AMD Ryzen 5, 16GB RAM, integrated GPU, 500GB SSD storage
Development Machine (Recommended)	Intel Core i7 / AMD Ryzen 7, 32GB RAM, dedicated GPU (NVIDIA GTX 1660 or higher, 6GB VRAM+), 1TB SSD storage
Model Training (Cloud/GPU)	Access to cloud GPUs such as NVIDIA Tesla T4, V100, or A100 with 16GB+ VRAM (via Google Colab Pro, AWS, or Kaggle)
Peripheral Devices	A standard HD webcam (minimum 720p, recommended 1080p) for capturing gestures

Table 3: Hardware Requirements

Software Requirements:

Programming Languages & Frameworks	Python, React Native, OpenCV, MediaPipe
Machine Learning & Deep Learning Tools	TensorFlow, Keras, CNN, LSTM
Datasets	RWTH-PHOENIX-Weather 2014T, ASLLVD (American Sign Language Lexicon Video Dataset), Custom-Collected Dataset
Architecture & Patterns	Microservice Architecture, Real-time Gesture Recognition Pipeline

Table 4: Software Requirements

1.14 Chapter Summary

This chapter outlined the background and challenges in sign language education for deaf and hard-of-hearing children, highlighting the lack of interactive tools with real-time feedback. The problem definition, statement, and motivation emphasized the need for a web-based platform that integrates gesture recognition and corrective feedback. Research objectives, gaps, contributions, and challenges were presented, along with the mapping of objectives to research questions and learning outcomes. Together, these elements establish the foundation for the proposed study, which will be expanded through the literature review in the next chapter.

CHAPTER 2: LITERATURE REVIEW

2.1 Chapter Overview

The literature review pertaining to the creation of an interactive web-based sign language learning platform is presented in this chapter. The problem domain is examined first, describing the difficulties deaf and hard-of-hearing (DHH) youngsters encounter in obtaining high-quality educational and communication materials. The use of computer vision, deep learning, and gesture identification in sign language learning is then examined, with an emphasis on current technical solutions. In order to promote engagement and retention, the evaluation also assesses child-centered design strategies and multimodal interaction techniques. The chapter concludes by outlining the shortcomings of the systems and research that are now available and pointing out the research gap that supports the necessity of the suggested remedy.

2.2 Problem Domain

2.2.1 Overview of Sign Language Learning

Sign language serves as the primary mode of communication for millions of deaf and hard-of-hearing (DHH) individuals across the world. However, access to structured sign language education remains severely constrained due to the shortage of qualified instructors, the absence of standardized curricula, and the limited availability of scalable, interactive learning tools (BeMyVoice, 2024; Dodandeniya et al., 2023). Traditional learning resources such as printed dictionaries, flashcards, and video demonstrations offer only basic vocabulary exposure and rely heavily on passive observation. These static methods lack interactivity, real-time assessment, and corrective feedback, which are essential for developing fluency and confidence (Deoghare et al., 2024; Milinda et al., 2024). Consequently, many DHH children can memorize individual gestures but struggle to apply them effectively for spelling, sentence formation, or everyday communication (Jayawardana et al., 2023; Fernando et al., 2025).

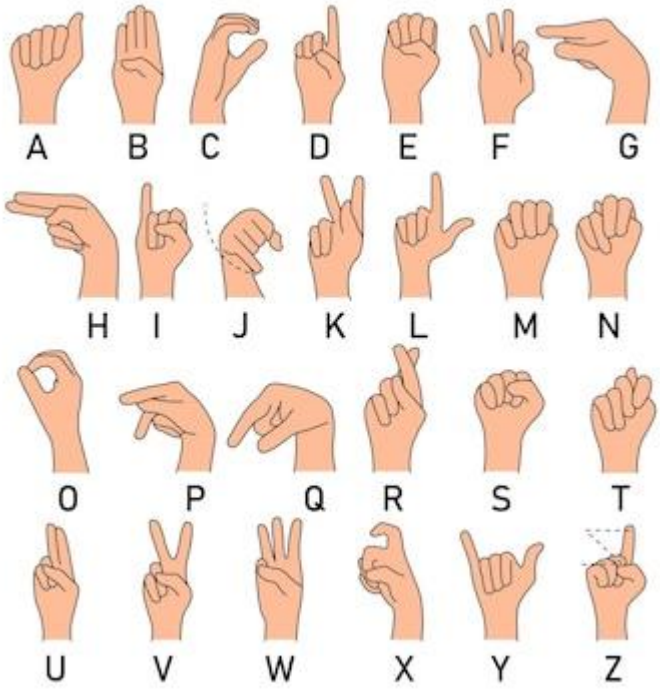

Figure 1: American Sign Language Alphabet

2.2.2 Importance of Interactive and AI-Driven Platforms

Recent developments in computer vision and artificial intelligence (AI) have opened up new avenues for revolutionizing sign language instruction. When combined with hand-tracking frameworks like MediaPipe, deep learning architectures like Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks have shown excellent accuracy in real-time sign gesture recognition. Personalized learning experiences are made possible by integrating these technologies into web-based platforms, which let students practice on their own while getting immediate performance tracking and remedial feedback. Additionally, gamification components like levels, progress indicators, and rewards increase motivation and long-term engagement, which makes learning more fun and efficient for kids. To close the ongoing gap between simple sign exposure and actual communicative mastery, these interactive and flexible methods are crucial (Fernando et al., 2025).

2.2.3 Application Areas

2.2.3.1 Educational Contexts

The proposed system can be integrated into special education classrooms, inclusive schools, and home-learning environments. By providing teachers with a supportive tool and enabling learners to practice independently, it helps overcome the scarcity of trained interpreters and personalized instruction (Dodandeniya et al., 2023).

2.2.3.2 Social and Communication Skills Development

Beyond classrooms, the platform prepares children for real-world communication. By moving bfrom letter-based practice to object-based dedicated signs, learners gain both literacy and functional vocabulary, supporting smoother interactions with peers, family, and society (BeMyVoice, 2024).

2.2.4 Challenges in Sign Language Learning Platforms

2.2.4.1 Engagement and User Experience

Children can lose interest if the platform is repetitive or overly technical. Designing a child-friendly, playful, and motivating interface is essential to ensure consistent use.

2.2.4.2 Scalability and Real-Time Recognition

Gesture recognition systems must perform accurately across different devices, camera qualities, and environmental conditions such as lighting or background clutter. Ensuring low latency while maintaining accuracy is critical for real-time learning (Milinda et al., 2024).

2.2.4.3 Accuracy and Robustness

Some hand gestures are visually similar (e.g., “M” vs. “N”), making them hard to distinguish. Models must be robust enough to handle variations in hand shapes, sizes, and signing styles without compromising recognition accuracy.

2.2.5 Proposed Architecture

The proposed solution is a web-based interactive sign language learning platform with six core layers:

1. Input Layer: Learners perform gestures via webcam.
2. Gesture Recognition Module: MediaPipe extracts hand landmarks; CNNs classify static letters; LSTMs/Temporal CNNs handle sequential spelling.
3. Feedback Engine: Real-time corrective cues (✓, hints, overlays).
4. Learning Modules:
    - Alphabet stage (A–Z).
    - Word-building stage (fingerspelling).
    - Vocabulary stage (dedicated object signs).
    - Mixed practice stage.
5. Gamification and Progress Tracking: Rewards, badges, and progress reports.
6. Web Interface: Accessible, child-friendly browser-based UI.

2.3 Existing Work

2.3.1 Vision-Based Hand/Gesture Recognition

Most systems rely on RGB video with hand-pose landmarks (e.g., MediaPipe) or image patches fed to CNNs for alphabet-level recognition. This approach enables real-time browser deployment and is common in learning tools focused on A–Z letters. Key limitation: strong results on static letters, but weaker performance for word-level or visually similar signs (e.g., M vs N, O vs S) (Milinda et al., 2024; Jayawardana et al., 2023).

2.3.2 Fingerspelling Temporal Dynamics

To recognize letter sequences, studies add temporal models (LSTM/TCN/CTC). These capture rhythm, hold time, and transition smoothness similar to timing signals in keystroke dynamics. Strengths: letter-to-word spelling (e.g., C-A-T) with error localization for feedback. Limitation: accuracy drops with inconsistent pace, occlusions, or camera angle changes (Sau et al., 2022; Fernando et al., 2025).

2.3.3 Wearables & Device-Assisted Signals

A smaller body of work attaches IMUs/EMG rings or uses depth cameras to stabilize recognition. Pros: robustness to lighting/background clutter. Cons: extra hardware reduces accessibility for classroom/home use and is outside the web-only scope of this project (BeMyVoice, 2024; Deoghare et al., 2024).

2.3.4 Full-Body/Upper-Body Pose

Some languages include non-manual markers (facial expression, head tilt, torso orientation). Pose-estimation and holistic landmark models improve word-level recognition but increase compute cost and complicate browser deployment (Jayawardana et al., 2023; Dodandeniya et al., 2023).

2.3.5 Multimodal Combinations

Recent systems pair hand landmarks with object images/prompts (to teach meaning) or lightweight text/phonology features. A few add captioning/AR contexts. Gap: very few platforms integrate a pedagogical flow that moves learners from alphabet → fingerspelling → dedicated object signs with instant corrective feedback in the web browser (Fernando et al., 2025; Deoghare et al., 2024).

2.4 Technological Review

The success of a web-based sign language learning platform depends heavily on the quality of datasets used to train recognition models and the preprocessing applied to ensure robustness in real-world use. Deep learning architectures such as CNNs and LSTMs require large amounts of diverse, high-quality training data. However, sign language datasets face challenges such as limited availability, variation across languages, and inconsistency in recording conditions. Preprocessing steps like normalization, scaling, and outlier removal are therefore critical to improve recognition accuracy and ensure the system can operate reliably in varied environments such as classrooms or homes.

2.4.1 Dataset and Preprocessing

2.4.1.1 Data Sources and Acquisition

A variety of datasets have been developed for sign language recognition:

- Alphabet-Level Datasets:

  Public datasets such as the ASL Alphabet Dataset (widely available via Kaggle) provide thousands of labeled images for 26 hand gestures representing letters. These datasets are widely used in CNN-based recognition studies.

- Word- and Sentence-Level Datasets:

  Datasets like RWTH-PHOENIX-Weather 2014T support continuous sign recognition and have been used for training sequence models such as LSTMs.

- Localized and Educational Datasets:

Tools like Visual Kids (Sri Lanka) demonstrate regional adaptations where datasets include alphabets, numbers, and objects for child-friendly applications. Such region-specific resources are valuable for contextual learning but are often small in scale.

- Custom Dataset Creation:

  Since sign language varies by country and context, custom datasets often need to be collected for specific projects. For this research, a curated dataset that includes static alphabet signs, fingerspelling sequences, and dedicated object-based signs will be essential to support scaffolded learning.

2.4.1.2 Preprocessing Techniques

Preprocessing transforms raw video or image data into clean, standardized inputs suitable for training deep learning models.

2.4.1.2.1 Noise Filtering and Normalization

- Noise Filtering: Removes background clutter, lighting inconsistencies, and jitter in hand detection. Hand landmark extraction libraries such as MediaPipe already provide smoothing filters to reduce noise.
- Normalization: Coordinates of hand landmarks are normalized relative to wrist position and palm size to ensure consistency across different users with varying hand sizes and distances from the camera.

2.4.1.2.2 Feature Scaling and Standardization

- Scaling ensures that extracted features such as joint distances, angles, or motion trajectories are within a fixed range, improving training convergence.
- Standardization (e.g., z-score normalization) aligns data distributions so that CNN or LSTM models treat all input features equally, reducing bias toward larger values.

2.4.1.2.3 Outlier Detection and Removal

- Outlier Detection: Identifies invalid frames, such as when a hand is partially outside the camera view or occluded.
- Removal or Correction: Such samples are either discarded or replaced with interpolated values to avoid introducing noise into the model.
- Outlier handling is particularly important for children's data, as they may perform signs inconsistently or with extra movements not present in standardized datasets.

2.5 Benchmarking and Evaluation

Benchmarking is essential for assessing the effectiveness of sign language recognition models and interactive learning platforms. While deep learning methods such as CNNs, RNNs, and hybrid multimodal systems have shown promising results, their evaluation must go beyond accuracy alone. Effective benchmarking considers not only recognition metrics but also computational efficiency, scalability, and the user experience in real-world learning contexts.

2.5.1 Evaluation Metrics for Authentication Models

2.5.1.1 Accuracy, Precision, and Recall

Accuracy, precision, and recall are the most widely used metrics in evaluating sign language recognition systems. Accuracy measures the proportion of correctly recognized gestures out of the total predictions, offering a broad indication of system performance. However, in gesture recognition tasks, accuracy alone can be misleading, especially when certain signs are more frequent in the dataset than others. Precision addresses this by evaluating how many gestures predicted as a specific class were correct, which is particularly important when dealing with visually similar signs such as *M* and *N*. Recall, on the other hand, measures the ability of the model to detect all actual instances of a sign in the dataset, ensuring that no learner attempts are overlooked. High recall is crucial in educational platforms, as missing a child's attempt to perform

a gesture may prevent timely feedback and hinder the learning process. Together, these three metrics provide a balanced assessment of recognition models, allowing researchers to determine whether systems are both reliable and responsive in real-world usage.

2.5.1.2 F1-Score and AUC-ROC

The F1-score and AUC-ROC are additional metrics that provide deeper insight into model performance beyond simple accuracy. The F1-score is the harmonic mean of precision and recall, making it particularly useful when evaluating systems where some classes are harder to recognize or when datasets are imbalanced. In sign language recognition, this is valuable because certain gestures, such as common vowels, may dominate training datasets, while less frequent signs may not be represented equally. The F1-score ensures that both precision and recall are weighted fairly, helping to avoid biased results. Similarly, the AUC-ROC (Area Under the Receiver Operating Characteristic Curve) provides a way to evaluate the trade-off between sensitivity and specificity across multiple classes. While less commonly applied in gesture recognition compared to accuracy or F1-score, AUC-ROC can be beneficial in assessing how well a system distinguishes between correct and incorrect gestures at different threshold levels. These metrics are therefore essential in providing a holistic view of performance, ensuring that recognition models are not only accurate in controlled datasets but also robust when deployed in real-world educational environments.

2.5.1.3 Computational Efficiency and Processing Time

For a sign language learning platform to be effective, recognition models must not only be accurate but also computationally efficient enough to operate in real time. In practice, this means minimizing latency so that learners receive immediate corrective feedback after performing a gesture. Studies suggest that delays greater than 100 milliseconds can disrupt user interaction and reduce engagement, especially in educational contexts where children expect instant responses. The ability to maintain high frame rates, typically measured in frames per second (FPS), is equally important for ensuring smooth operation across different browsers and hardware configurations. While CNN-based models are generally lightweight and well-suited for fast inference, more complex architectures such as LSTM or hybrid CNN–LSTM networks can introduce higher computational costs, potentially slowing responsiveness. Balancing recognition accuracy with

processing time is therefore a key benchmark for evaluating system performance, as prolonged delays in recognition or feedback may discourage consistent learning and reduce overall usability.

2.5.2 Comparative Analysis of Existing Models

2.5.2.1 Benchmarking Studies

Benchmarking studies on sign language recognition systems reveal that a wide range of models have been developed, each with unique trade-offs in performance. Convolutional Neural Networks (CNNs) remain the most common choice for static gesture recognition, achieving recognition accuracies between 85% and 95% on alphabet datasets such as ASL. Their strength lies in their ability to extract spatial features efficiently, making them lightweight and suitable for browser-based or real-time deployment. However, CNNs alone struggle with sequential gestures, as they cannot capture the temporal dynamics of fingerspelling or multi-frame signs. To address this limitation, researchers have integrated Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) models, which excel in capturing sequence information and transitions between letters. These hybrid CNN–LSTM systems improve recognition of fingerspelling sequences but typically demand larger datasets and greater computational resources, limiting their deployment in resource-constrained environments. More recent work has also explored multimodal and holistic approaches, integrating hand landmarks, facial expressions, or body pose to enhance accuracy. While these methods show promise, they often introduce additional complexity and higher latency, which can compromise real-time performance in web applications.

2.5.2.2 Real-World Performance and User Experience

Although many models report high accuracy in benchmarking studies, real-world performance often reveals significant gaps between laboratory results and practical usability. Systems tested in controlled environments with uniform lighting, clear backgrounds, and fixed cameras consistently demonstrate strong recognition rates. However, when deployed in classrooms or home-learning contexts, performance is affected by environmental variability such as poor lighting, cluttered backgrounds, and inconsistent camera positioning. Moreover, children's gestures are less consistent than adults', with variations in hand orientation, speed, and precision that further challenge recognition models. In terms of user experience, responsiveness and feedback

quality play a critical role. Learners are more likely to engage with platforms that provide immediate and encouraging feedback rather than delayed or overly technical responses. Studies on gamified learning systems show that progress tracking, rewards, and interactive challenges significantly improve engagement but these systems often lack robust AI-driven feedback. Thus, while recognition models perform well under controlled conditions, their real-world educational impact depends equally on system robustness, interface design, and motivational features.

2.5.3 Challenges in Benchmarking Multimodal Systems

Benchmarking multimodal systems for sign language recognition presents several unique challenges. First, there is a lack of standardized datasets that include multiple modalities such as hand landmarks, facial expressions, and contextual objects. This makes direct comparison between models difficult, as different studies train and evaluate on non-uniform datasets. Second, the diversity of sign languages creates inconsistencies in benchmarks, since gestures differ across

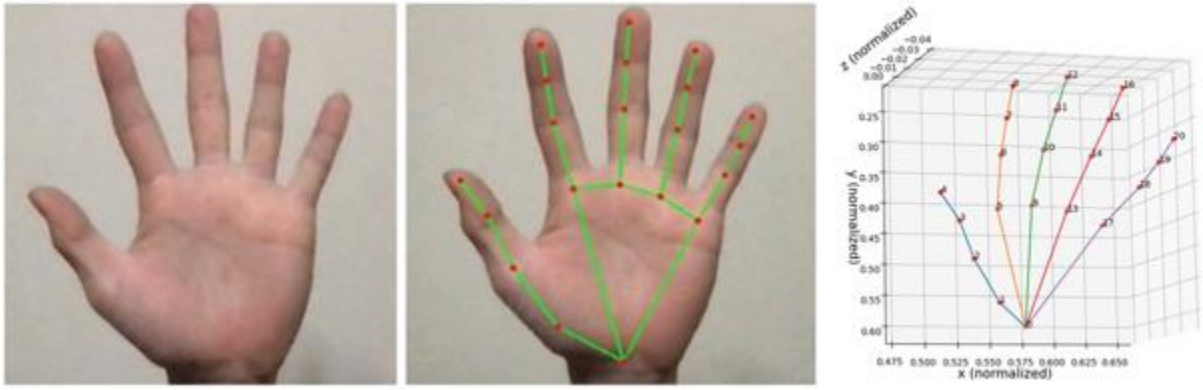

Figure 2: Hand pose estimation using media pipe

ASL, ISL, BSL, and local variants such as Sri Lankan Sign Language. Third, hardware and deployment environments influence results significantly: systems optimized for GPUs or high-performance desktops may not perform reliably on standard laptops or browsers used in classrooms. Another challenge lies in user variability children's hand sizes, signing speeds, and styles differ greatly from adults, yet most datasets are adult-centric, creating a mismatch between training data and target users. Finally, few studies evaluate usability metrics such as learner engagement, motivation, and long-term retention, even though these are critical in educational

contexts. Together, these challenges underline the importance of not only benchmarking recognition accuracy but also evaluating multimodal systems for scalability, inclusiveness, and child-cantered usability.

2.6 Chapter Summary

This chapter reviewed literature on sign language learning and recognition, focusing on the role of AI-driven platforms in addressing the educational challenges faced by deaf and hard-of-hearing (DHH) children. Existing research demonstrates that CNNs and LSTMs achieve strong recognition accuracy for static alphabets and fingerspelling, while gamification improves engagement and motivation. However, most systems remain limited to alphabets, lack scaffolded pathways to word-building and vocabulary, or rely on controlled datasets that do not generalize well to real-world environments. Wearable and multimodal approaches offer improvements in robustness but are less suitable for accessible web-based deployments.

The technological review highlighted available datasets, preprocessing techniques, and benchmarking methods. While ASL datasets and hybrid CNN–LSTM models show promise, the lack of diverse, child-focused, and region-specific data restricts scalability. Benchmarking studies primarily focus on accuracy, with less emphasis on latency, robustness, or user experience, all of which are critical for educational contexts. These gaps establish the relevance of the proposed research: a web-based interactive learning platform that combines real-time gesture recognition, corrective feedback, and gamified progression to support literacy, vocabulary development, and inclusive communication for DHH children.

CHAPTER 03: METHODOLOGY

3.1 Chapter Overview

This chapter outlines the research, development, and project management methodologies adopted for this study. It also details the project schedule, deliverables, required resources, data requirements, risks, and mitigation strategies. The methodologies were carefully selected to ensure an efficient and structured approach to achieving the research objectives.

3.2 Research Methodology (Saunders' Research Onion in Table Form)

Layer	Choice	Justification
Philosophy	Pragmatism (both quantitative & qualitative)	Quantitative data is required for training/testing gesture recognition models, while qualitative data (e.g., user feedback, teacher observations) helps evaluate usability and engagement.
Approach	Deductive	The research tests the effectiveness of CNN- and landmark-based models for sign recognition in educational contexts. Hypotheses about recognition accuracy, latency, and user learning outcomes are validated experimentally.

Layer	Choice	Justification
Methodological Choice	Mixed Methods	Combines quantitative metrics (accuracy, precision, recall, F1-score, latency) with qualitative evaluations (user surveys, interviews with educators).
Strategy	Experimental, Surveys, Observation, Prototyping	Controlled experiments to test recognition accuracy, surveys/interviews to gather learner and teacher feedback, and observation during trials to study engagement and usability.
Time Horizon	Cross-sectional	Data collection and evaluation will occur at defined project phases (alphabet stage, word-building stage, vocabulary stage) rather than over extended longitudinal use.
Data Collection	Secondary + Custom Data	Public datasets (ASL Alphabet, WLASL subsets, LSA64) are used for training/testing. A small custom dataset is collected from local learners to adapt to real-world usage

Table 5: Research Methodology (Saunders' Research Onion in Table Form)

3.3 Development Methodology

The project adopts a hybrid approach combining Prototyping and Agile. Prototyping allows iterative development and testing of recognition models and user interfaces in stages alphabet recognition, word-building, and vocabulary expansion ensuring continuous refinement based on feedback and performance. Agile structures these prototypes into sprints, aligning with key milestones such as Proposal, Proof of Concept, Prototype, Demo, and MVP. This approach ensures flexibility to adapt models (e.g., CNN, CNN–LSTM) and features as needed, while maintaining steady progress. The combination of Prototyping and Agile is particularly suited for AI-based educational platforms, as it supports both technical experimentation and user-cantered design, directly addressing the project's research gap.

3.3.1 Requirement Elicitation Methodology

Technique	Purpose
Literature Review	Identify gaps in sign language systems and define technical requirements such as real-time recognition and corrective feedback
Interviews	Gather insights from special education teachers on effective teaching strategies and child-friendly features.
Surveys	Collect feedback from potential users (learners, teachers, parents) about usability and engagement.
Document Analysis	Study existing sign language learning applications to identify strengths and limitations.
Brainstorming	Generate innovative features such as gamification, progress tracking, and dual practice modes (fingerspelling + vocabulary).
Observation	Monitor how children interact with existing static tools to refine interaction design.

Prototyping	Validate design choices and gesture recognition accuracy iteratively.
Self-Evaluation	Ensure project outcomes remain aligned with research objectives and deliverables.

Table 6: Requirement Elicitation Methodology

3.3.2 Design Methodology

SSADM is used as the design approach for this project, providing a structured method to analyse and design the gesture recognition workflows and learning interactions of the system. For the sign language learning platform, SSADM ensures clarity and efficiency through well-defined phases, including requirements analysis, data flow modelling, and feedback mechanism design. Techniques such as Data Flow Diagrams (DFDs) and Entity-Relationship Diagrams (ERDs) are applied to visualize system processes, while UML Use Case Models represent interactions between learners, teachers, and the platform. This structured approach enhances system documentation, supports stakeholder involvement, and ensures the platform design remains resilient, scalable, and user cantered.

3.3.3 Programming Paradigm

To ensure modularity, maintainability, and efficiency across the different components of the system, a combination of programming paradigms has been adopted based on the nature of the task:

- Object-Oriented Programming (OOP): OOP was employed for the core system architecture using Python and JavaScript. This paradigm supports modular design through encapsulation and reusability, enabling independent development of core modules such as the gesture recognition engine, feedback system, and gamified interface. The modularity also allows future extensions, such as scaling from alphabet-level recognition to word and sentence recognition, which has been highlighted as a necessary feature in interactive learning systems (Deoghare et al., 2024; Jayawardana et al., 2023).

- Structured Programming: Procedural subroutines were applied to tasks such as dataset loading, preprocessing, and evaluation routines. This approach ensures clear control flow, making linear tasks like normalization, augmentation, and logging efficient and less error-prone. Structured programming has been applied successfully in prior real-time sign language recognition systems to streamline computational tasks (Milinda et al., 2024; Jayasuriya et al., 2020).
- Functional Programming**:** Elements of functional programming, such as pure functions and immutability, were applied in data transformation processes. Examples include landmark normalization, augmentation operations, and preprocessing pipelines. While not used as the primary paradigm, functional programming improves predictability, simplifies debugging, and supports reusability in data-centric computations (Sau et al., 2022; Fernando et al., 2025).

3.3.4 Testing Methodology

Testing was conducted in two phases to evaluate both the technical performance of the recognition models and the usability of the integrated platform.

- **Model Testing:** The recognition models, including CNN, LSTM, and hybrid CNN–LSTM architectures, were evaluated using standard performance metrics such as accuracy, precision, recall, F1-score, and latency. Datasets such as the ASL Alphabet dataset and RWTH-PHOENIX-Weather 2014T were used for training and validation, while data augmentation simulated real-world conditions like lighting changes, hand orientation, and background clutter (Jayawardana et al., 2023; Milinda et al., 2024). CNN models generally performed well for static alphabet signs, whereas LSTM and hybrid approaches were required for effective fingerspelling recognition, consistent with findings in recent studies (Fernando et al., 2025; Sau et al., 2022).
- **Prototype Testing:** The full platform integrating gesture recognition, corrective feedback, gamification, and the user interface was tested with learners and teachers in controlled sessions. This phase assessed system responsiveness, feedback effectiveness, and learner engagement. Prior research has demonstrated that features such as gamification and

progress tracking enhance motivation in DHH children, while immediate corrective feedback improves learning outcomes (Dodandeniya et al., 2023; Deoghare et al., 2024). Prototype testing also considered scalability and accessibility across browsers and hardware devices, an issue highlighted in studies on web-based sign recognition systems (Jayasuriya et al., 2020; Jayawardana et al., 2023).

3.3.5 Solution Methodology (Expanded Subsections)

A structured pipeline was used in the creation of the web-based interactive sign language learning platform, guaranteeing methodical advancement from data gathering to model improvement:

1. Dataset Collection: To train gesture recognition models, publicly accessible datasets like the ASL Alphabet dataset and RWTH-PHOENIX-Weather 2014T were utilized. Furthermore, for context-specific adaptation, smaller child-centered datasets like Visual Kids were examined.

2. Data preprocessing: To increase the generalizability of the model, raw images and video sequences were processed. To replicate various real-world scenarios, steps included normalizing hand landmarks, reducing noise, and using augmentation techniques like flipping, rotation, and brightness adjustment (Milinda et al., 2024; Fernando et al., 2025).

3. Feature Engineering and Selection: MediaPipe was used to extract important features from pose data and hand landmarks. Spatial coordinates, inter-joint distances, angular relationships, and temporal motion sequences were among the features used to differentiate between dynamic fingerspelling and static alphabet signs (Sau et al., 2022; Deoghare et al., 2024).

4. Model Selection: Because CNN-based architecture performed well in image classification tasks, they were used for static alphabet recognition (Jayasuriya et al., 2020; Milinda et al., 2024). As suggested by earlier research on sign recognition, LSTM and hybrid CNN–LSTM networks were used for sequential data to capture temporal dependencies in fingerspelling (Sau et al., 2022; Fernando et al., 2025).

5. Model Training: TensorFlow and PyTorch frameworks were used for training, and optimization strategies like cross-validation and dropout regularization were used to avoid overfitting. The models were trained gradually, beginning with alphabets and moving on to vocabulary signs and word-building (Deoghare et al., 2024).

6. Testing: Metrics including accuracy, precision, recall, F1-score, and computational efficiency were used to assess the models. In order to verify system robustness, testing took into account both benchmark datasets and simulated classroom environments (Jayawardana et al., 2023; Dodandeniya et al., 2023).

7. Feedback Loop: Model hyperparameters, preprocessing techniques, and UI/UX features were iteratively improved based on testing and prototype evaluation results. This made sure that throughout development cycles, learner usability and technical recognition accuracy increased steadily (Fernando et al., 2025; Deoghare et al., 2024).

3.4 Project Management Methodology

3.4.1 Schedule

3.4.1.1 Gantt Chart

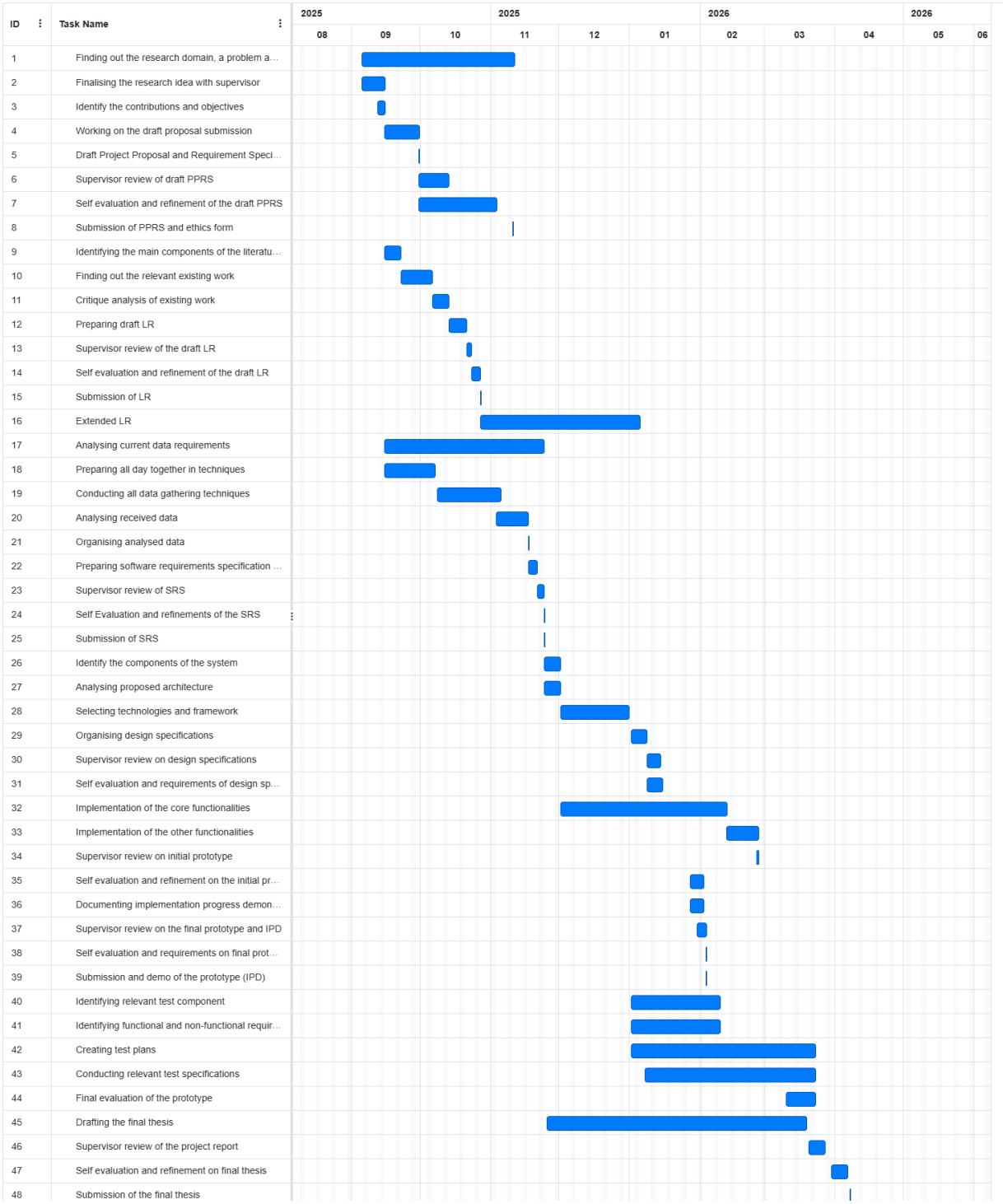

Figure 3 : Gantt Chart

3.4.1.2 Deliverables: Literature review, prototype development, final report submission.

Delivarebles	**Dates**
Project Proposal	
Literature Review	
Project Proposal	10th Oct 2025
Project Proposal and Requirement Specification – final draft	
Project Proposal and Requirement Specification – Final	06th Nov 2025
Design Document	
Prototype	2nd Feb 2026
Interim Project Demo	2nd Feb 2026
Implementation	2nd Feb 2026
Testing	
Evaluation	
Thesis Submission	30th Mar 2026
Prototype	30th Mar 2026
Dataset and Source Code	30th Mar 2026

Table 7: Deliverables

3.4.2 Resource Requirements

3.4.2.1 Data

The project requires publicly available large-scale sign language datasets, including the ASL Alphabet dataset, RWTH-PHOENIX-Weather 2014T for continuous signs, and child-focused datasets such as Visual Kids. Additional data will be collected or augmented to improve robustness in varied real-world learning contexts.

3.4.2.2 Skills

Key technical skills required for this project include machine learning and deep learning model development, Python programming, data preprocessing, and web-based system design. Knowledge of libraries such as TensorFlow, PyTorch, OpenCV, and MediaPipe is essential for gesture recognition, while expertise in front-end web frameworks supports user interface development.

3.4.3 Risk and Mitigation (Expanded with Severity and Frequency)

A risk assessment was carried out to identify potential challenges during development and deployment. Each risk was rated based on severity and frequency, with mitigation strategies proposed.

Risk	**Severity**	**Frequency**	**Mitigation Strategy**
Insufficient training data	5	3	Use data augmentation techniques or source additional datasets.
Poor model accuracy	4	3	Optimize model parameters and incorporate hybrid CNN–LSTM structures.
Overfitting	5	4	Apply regularization, cross-validation, and expand dataset size.

Risk	Severity	Frequency	Mitigation Strategy
Hardware limitations	3	2	Use cloud-based GPU resources such as Google Colab Pro or AWS.
Real-time latency	4	3	Optimize model inference pipeline and compress trained models.
Stakeholder dissatisfaction	3	3	Conduct regular feedback sessions with teachers and test users.
Accessibility issues	3	2	Design responsive interfaces and test across multiple browsers.
Data privacy and ethical issues	2	2	Collect minimal personal data and comply with ethical guidelines.

Table 8: Risk and Mitigation (Expanded with Severity and Frequency)

3.5 Chapter Summary

This chapter outlined the chosen research, development, and project management methodologies, along with project scheduling, required resources, and potential risks. These selections ensure a structured and systematic approach to achieving the research objectives.

CHAPTER 04: SOFTWARE REQUIREMENT SPECIFICATION (SRS)

4.1 Chapter Overview

This chapter outlines the Software Requirement Specification (SRS) for the AI-driven, web-based interactive sign language learning platform. It defines the system's purpose, requirements, stakeholders, and interactions through diagrams and structured analysis. The SRS identifies both functional and non-functional requirements, ensuring that user and technical expectations are clearly defined before system development. It also employs the MoSCoW prioritization technique to determine which features are essential, desirable, or optional for successful implementation.

4.2 Rich Picture Diagram

The Rich Picture Diagram presents the overall structure and stakeholder interactions within the proposed AI-driven web-based interactive sign language learning platform. At its core, the system integrates CNN–LSTM and MediaPipe-based hand tracking to recognize gestures in real time and provide immediate corrective feedback. Deaf and hard-of-hearing (DHH) learners interact with the platform through a webcam, while teachers upload lessons and monitor progress. Administrators manage the system, ensuring performance and accessibility, and parents can view learners' progress reports to support home-based learning.

All data, including gesture inputs and performance analytics, are stored in a cloud-based database, allowing continuous refinement and scalability. The web interface ensures accessibility across different devices, removing the need for specialized hardware. Gamified elements and adaptive feedback loops enhance motivation and retention, creating an engaging and inclusive learning experience (Milinda et al., 2024; Fernando et al., 2025; Deoghare et al., 2024).

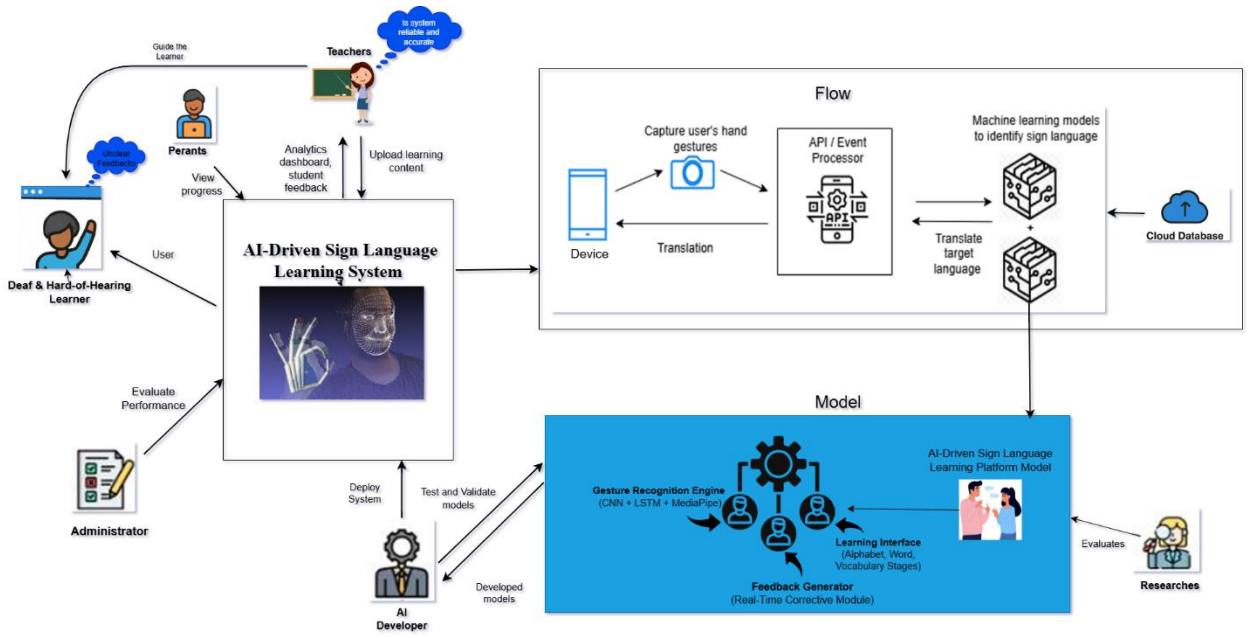

Figure 4: Rich Picture Diagram

4.3 Stakeholder Analysis

The stakeholders involved in the AI-Driven Web-Based Interactive Sign Language Learning Platform are depicted using a Stakeholder Onion Model, which illustrates the varying levels of interaction and influence each group has on the system. The model places the AI-powered learning platform at the centre, surrounded by internal and external stakeholders who contribute to its operation, development, and educational impact.

4.3.1 Stakeholder Onion Model

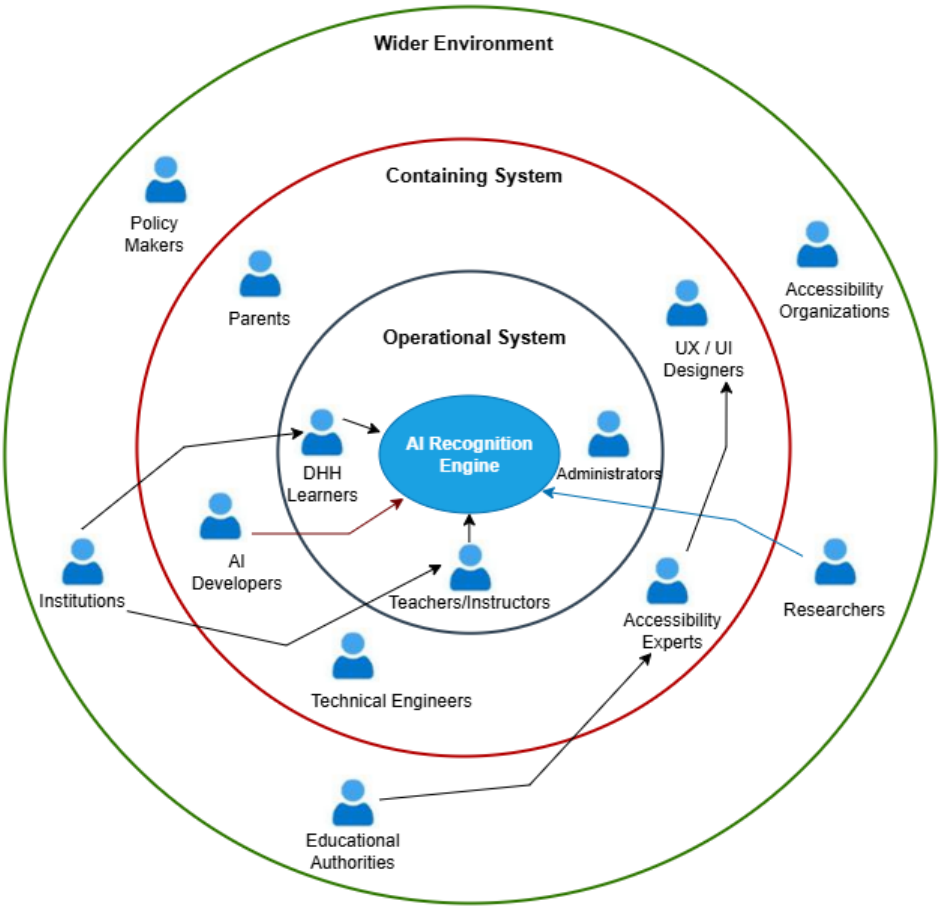

Figure 5: Stakeholder Onion Model

4.3.2 Stakeholder Viewpoints

Stakeholder	Role	Description
		Operational System Stakeholders
DHH Learners	Primary Users	Interact with the system through webcam-based gesture inputs to learn and practice sign language. They receive real-time corrective feedback from the AI model.
Teachers, Instructors	Content Creators	Upload lessons, monitor learner performance, and evaluate progress. They play a critical role in maintaining educational quality and relevance.
Administrators	System Managers	Manage platform operations, user accounts, and content databases. They ensure that the system remains functional, secure, and user-friendly.
		Containing System Stakeholders
Parents / Guardians	Supportive Observers	Monitor learner progress through performance reports and encourage consistent engagement outside school settings.
AI Developers / Technical Engineers	System Developers	Develop and maintain the CNN–LSTM recognition model, improve system accuracy, and ensure real-time performance on web platforms.
Accessibility Experts	Compliance Advisors	Ensure that the interface adheres to accessibility standards (e.g., WCAG 2.1) and accommodates diverse user needs, including visual and motor impairments.
UX / UI Designers	Design Specialists	Optimize the platform's visual layout and interactivity to suit children's learning styles and improve engagement.

Stakeholders in Wider Environment		
Educational Authorities	Regulators	Provide guidelines for implementing assistive learning technologies within educational curricula and ensure compliance with national standards.
Policy Makers	Strategic Decision-Makers	Formulate inclusion policies and allocate funding for digital learning initiatives that support students with disabilities.
Research Collaborators / Academic Institutions	Innovation Partners	Conduct ongoing studies to improve AI model accuracy, expand sign datasets, and validate the platform's educational effectiveness.
Accessibility Organizations	Advocates	Promote inclusive digital education and ensure equal learning opportunities for DHH children in alignment with global accessibility goals.

Table 9: Stakeholder Viewpoints

4.4 Selection of Requirement Elicitation Methodologies

Requirement elicitation is a vital phase in developing the AI driven sign language learning platform, as it ensures that user needs especially those of deaf and hard-of-hearing individuals, are properly understood and translated into actionable system requirements. The selected techniques are outlined below.

Technique 1 – Literature Review
The literature review presented a solid theoretical background, in that the existing limitations of sign language learning technologies have been identified, and the best practices in gesture recognition and inclusive design have been outlined. By examining the existing research the key research gaps could be identified, including the lack of web-based real-time feedback and a lack of interactivity that were child-centred, which directly influenced the system purpose and functionality scope.

Technique 2 – Observations
Teachers and students in special-education classrooms had their observations to get to know how sign language is taught and practiced today. This enabled to recognition the real-life difficulties, including less time working with the teacher, the lack of interactive resources, and the inattention of children. These things guided the design of intuitive interfaces, short practice sessions and gamified learning activities to enhance engagement.
Technique 3 – Interviews
Teachers of the deaf, speech therapists, experts of accessibility and parents were also interviewed semi-structured. The objective was to collect various opinions regarding system usability, accessibility, and pedagogical congruity. The insights emphasized the importance of correct visual feedback, low-latency recognition, and the inclusion of both fingerspelling and vocabulary-level signs.
Technique 4 – Prototyping
An early low fidelity prototype was designed to check user interaction patterns and the gesture-recognition and feedback modules were then developed by refinement. Every version was also tested on a small group of users in order to test the responsiveness of the system and the effectiveness of learning. Prototyping also created feasible evidence to enhance the accuracy of a model, user experience, and the ability to confirm that hybrid CNN-LSTM architecture could run effectively within a browser-based platform (Sau et al., 2022).
Technique 5 – Brainstorming
Creative feature development was enabled through brainstorming with mentors of the project, artificial intelligence developers, and educators. Such sessions resulted in the introduction of the gamification factor like badges, point-based systems, and interactive quizzes and the thought of broadening the curriculum beyond the alphabet recognition to object-based sign learning. Brainstorming was used to ensure that the pedagogical and technical views were pulled together in system design.

Technique 6 – Self-Evaluation
The self-evaluation was an aspect that was incorporated in the entire project lifecycle to provide some form of consistency in the objectives, deliverables, and expectations of the stakeholders. Periodic review of progress evaluated the performance of the model, the accessibility of UI, and the feedback provided to the learners and enabled the research team to implement changes. This reflection process increased reliability and inclusiveness of the system.

Table 10: Selection of Requirement Elicitation Methodologies

4.5 Discussion of Findings

4.5.1 Literature Review Findings

Citation	**Findings**
(Deoghare et al., 2023)	Developed *An Interactive Sign Language-Based Mobile Application for Deaf People* using CNN-based recognition. While effective for static alphabet gestures, the system lacked real-time feedback and web compatibility, limiting accessibility for classroom and home use.
(Jayawardana et al., 2024)	Introduced *BeMyVoice*, a communicative platform assisting deaf students through sign recognition and translation. Although it improved accessibility, the system emphasized communication support rather than progressive sign learning or gesture correction for education.
(Fernando et al., 2024)	Proposed *Interactive Sign Language Learning System Using Computer Vision and Deep Learning*, achieving real-time recognition via CNN-LSTM. However, it did not incorporate adaptive learning or engagement mechanisms such as gamification for child learners.
(Jayasuriya et al., 2020)	Created *i-Sign: Sign Language Learning Application via Gamification*, focusing on engagement through game-based learning. Despite high motivational value, the model lacked AI-driven feedback and real-time gesture validation.

(Sau et al., 2022)	Designed an *ASL Recognition Model using Recurrent Neural Networks (RNNs)* for continuous gesture detection. The model demonstrated robust sequential recognition but was not optimized for browser-based, low-latency operation suitable for inclusive education settings.

Table 11: Literature Review Findings

4.5.2 Interviews Findings

Interviews were conducted with key stakeholders including teachers, DHH learners, and AI developers to gain deeper insights into the challenges, needs, and expectations regarding the interactive sign language learning platform.

Codes	**Themes**	**Conclusion**
Real-time Feedback	Learning Engagement and Accuracy	Participants consistently emphasized that most existing sign language learning tools still depend on *static teaching methods* such as dictionaries, flashcards, and pre-recorded videos. While these materials provide basic vocabulary exposure, they lack interactive features that enable learners to practice signs dynamically. Teachers and DHH learners agreed that *real-time corrective feedback* would greatly enhance engagement, accuracy, and motivation highlighting the urgent need for an interactive AI-driven platform.

Gesture Recognition Accuracy	System Performance and Usability	Teachers and developers noted that the system should handle diverse signing styles (e.g., hand size, speed) for robust performance across different DHH learners. Real-time accuracy is crucial to avoid learning delays.
Interface Simplicity	Accessibility and Child-Friendliness	Interviews revealed a strong preference for simple, visual-based interfaces with minimal text. Stakeholders suggested using clear icons and visual aids to enhance accessibility for young learners.
Progress Tracking	Motivation and Parental Engagement	Parents and teachers both stressed the importance of progress-tracking features, allowing them to track improvements and areas for support.

Table 12: Interviews Findings

4.5.3 Survey Findings

Question / Area	**Summary of Key Findings**
Participant Background	A majority (60%) of them were familiar with the sign language (although not everyone) as teachers or pupils, whereas 40% of them were not involved directly in the field but were aware of it.
Familiarity with Existing Tools	Mostly people were slightly acquainted with flashcards, dictionaries, and YouTube tutorials. They were plain and interactional.
Effectiveness of Current Systems	80% considered existing tools as either moderate or inactive, which proved their insignificance in active learning.

Identified Problems in Current Methods	Inadequate interactivity and absence of real time feedback were noted by all the participants as a big problem. Low engagement and poor comprehension were some of the secondary issues.
Preferred Platform Features	Real-time recognition and feedback of gestures, interactive games, tracking the progress, and demonstrations became the most sought-after features.
System Accessibility	It was also noted that all participants would like the system to be based on any web browser and not necessarily be executed via mobile-based apps and special hardware, which is inclusive.
AI in Learning	80% of the participants had a strong agreement that AI can simplify and enhance the process of learning sign language.
Comfort with Technology	Almost every respondent said that they were comfortable with digital learning platforms.
Interest in Future Participation	Most of them were not reluctant to test or revise the developed system at the later development phases of the research.

Table 13: Survey Findings

4.5.4 Brainstorming/Observation

4.5.4 Brainstorming/Observation

Criteria

The brainstorming and observation process was to find new innovative ideas on how to improve interactive sign language learning using AI and deep learning. These sessions consisted of interviews with domain experts, deaf and hard-of-hearing (DHH) teacher, and developer. The key goals were to resolve the problem of engagement with the tools of the static sign language, to introduce the feedback of real-time gestures, and to make choices of the design that being child-friendly and inclusive. Interactional observation of the children with already available learning applications was also done to optimize the usability, responsiveness, and engagement aspects.

Findings

• **Interactivity and Engagement:** Scholars stressed that children become easily bored with passive systems of learning like videos or flashcards. It was suggested to use gamification features such as badges, progress level, and rewards to continue motivation and enhance learning results.

• **Real-Time Feedback:** Teachers emphasized the significance of real-time corrective feedback in practicing the gesture so that the understanding and memorization can be enhanced. It was observed that delayed feedback demoralizes learners and causes decreased accuracy.

• **System Accessibility:** The respondents emphasized the importance of a web-based application that is compatible with a wide range of browsers and devices and does not require the use of mobile applications or expensive equipment

• **Child-Centered Interface:** The brainstorming activities focused on creating the colourful and intuitive and icon-based interface with little text to accommodate young learners and non-readers.

• **Recognition Model Requirements:** The developers came to the conclusion that a hybrid CNNLSTM system would be appropriate to process both stationary alphabet signs and movable fingerspelling sequences with the preservation of real-time.

4.6 Summary of Findings

Id	Finding	Literature Review	Interviews	Brainstorming/ Observation	Survey
1	Lack of Real-Time Feedback in Existing Systems	✓	✓	✓	✓
2	Need for Interactive and Engaging Learning Tools (e.g., Gamification)	✓		✓	✓
3	Importance of Real-Time Gesture Recognition for Effective Learning	✓	✓	✓	
4	Preference for Simple, Visual-Based User Interfaces		✓	✓	✓
5	Demand for Scalable, Web-Based Platforms	✓		✓	✓
6	Necessity of Adaptive Learning Systems to Accommodate Diverse Signing Styles	✓	✓	✓	
7	Need for Progress Tracking to Support Motivation and Parental Involvement	✓	✓		✓
8	AI Can Make Sign Language Learning More Effective and Accessible	✓		✓	✓

9	Challenges in Current Systems: Static, Non-Interactive Design	✓	✓		✓
10	Importance of Localized Sign Language Content for Inclusivity	✓		✓	

Table 14: Summary of Findings

4.7 Context Diagram

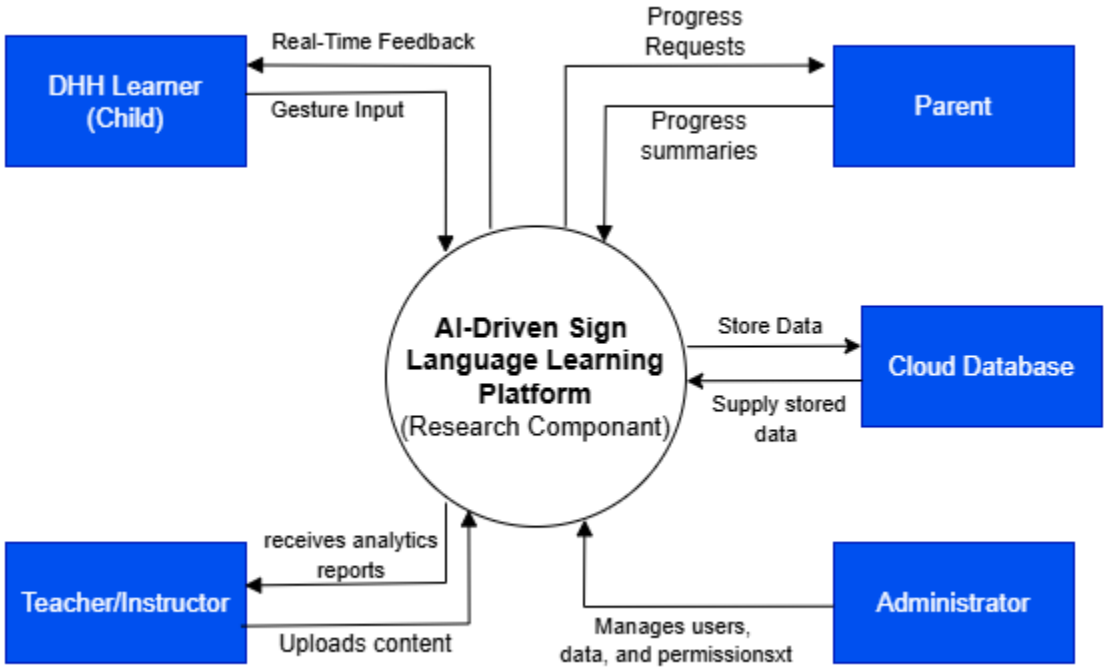

Figure 6: Context Diagram

4.8 Use Case Diagram

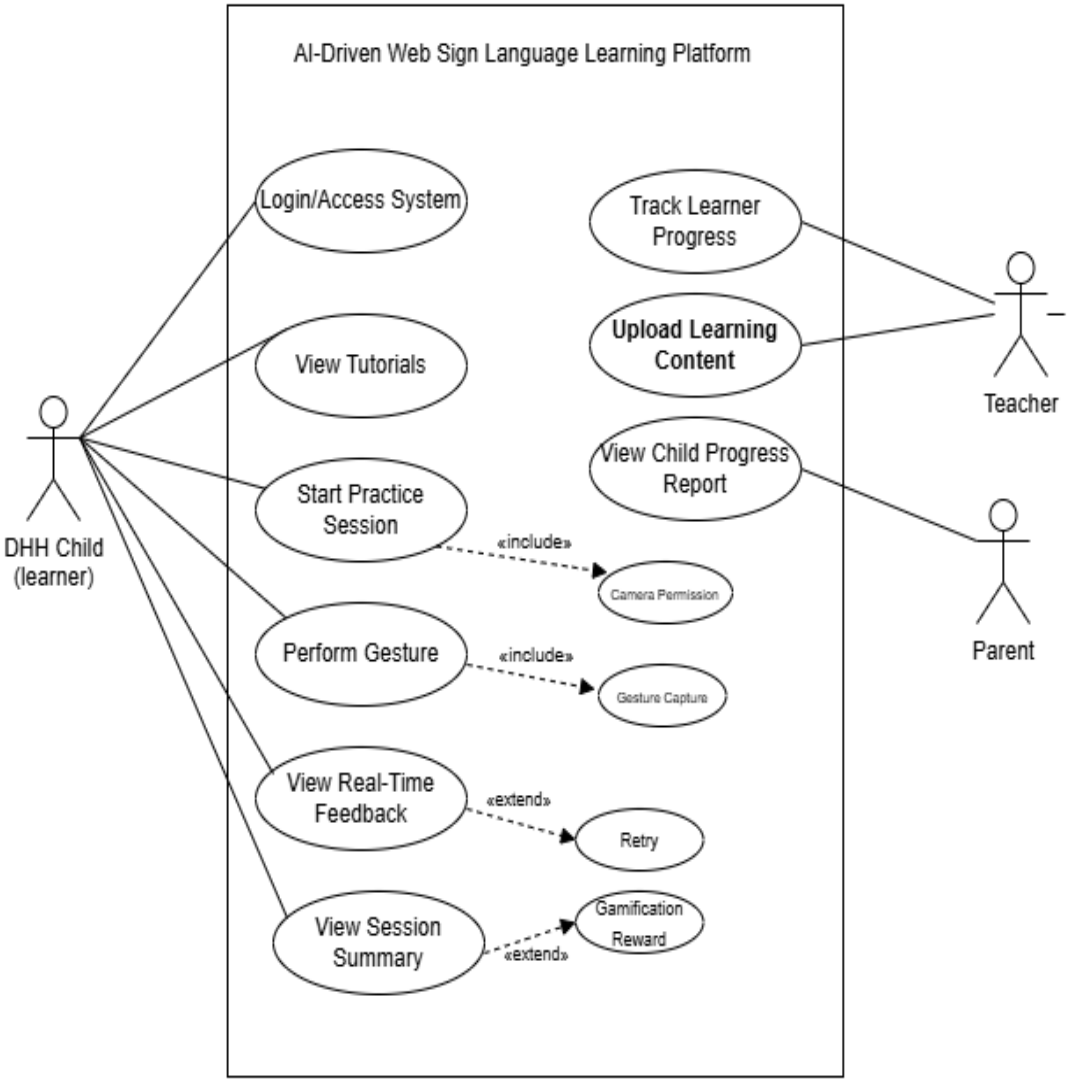

Figure 7: Use Case Diagram

4.9 Use Case Descriptions

Field	Description
Use case name	Login / Access System
Description	Authenticates the user (learner/teacher/parent) and loads the correct role-based dashboard.
Actors	DHH Child (Learner), Teacher, Parent, System
Preconditions	User has an account (or guardian-assisted account). Internet connection available.
Trigger	User submits credentials/OTP.
Main flow	**Actor:** enters credentials → **System:** validates identity, sets role/session, routes to dashboard.
Exception flow	Invalid credentials; locked account; consent not verified → show message and recovery options.
Postconditions	Authenticated session established with role permissions.

Table 15: Use Case Descriptions 1

Field	Description
Use case name	View Tutorials / Demos
Description	Learner watches a short demonstration (video/animation) that shows the handshape, orientation, and movement for a target letter/number/word.
Actors	DHH Child (Learner), System
Preconditions	Learner is logged in; lesson content is published.
Trigger	Learner selects a lesson or taps “Show Demo”.

Main flow	**Actor:** selects item → **System:** loads and plays demo with visual cues; learner proceeds to practice.
Exception flow	Media not available/slow network → provide low-bandwidth fallback or static cues.
Postconditions	Learner is prepared to attempt the gesture.

Table 16: Use Case Descriptions 2

Field	**Description**
Use case name	Start Practice Session
Description	Initializes a practice session for a chosen set (letters/numbers/words), prepares the capture pipeline, and displays the current target.
Actors	DHH Child (Learner), System
Preconditions	Learner is logged in; a practice set is selected.
Trigger	Learner clicks “Start Practice”.
Main flow	**System:** displays target → requests camera permission → opens webcam → brief hand/framing check → shows “Ready” state.
Exception flow	Camera blocked/unsupported; low light/background clutter → show guidance and retry.
Postconditions	Session active; system ready to receive gesture attempts.

Table 17: Use Case Descriptions 3

Field	Description
Use case name	Perform Gesture
Description	Learner performs the target sign in front of the webcam; system captures frames and runs recognition.
Actors	DHH Child (Learner), System
Preconditions	Practice session active; camera streaming.
Trigger	Learner positions hand and performs the sign.
Main flow	**System:** detects hand → extracts landmarks → classifies (CNN for static; LSTM/TCN for sequences) → stores score and timestamp.
Exception flow	Hand out of frame/occluded → prompt to re-center; environment warnings if needed.
Postconditions	Attempt recorded with predicted label, confidence, and latency.

Table 18: Use Case Descriptions 4

Field	Description
Use case name	View Real-Time Feedback
Description	Displays immediate, child-friendly feedback ("Correct!", "Almost—straighten fingers") and offers retry/next actions.
Actors	DHH Child (Learner), System
Preconditions	At least one attempt has been inferred.
Trigger	System finishes inference for the attempt.
Main flow	**System:** maps errors to tips/overlays; updates accuracy/streaks; **Actor:** chooses "Try Again" or "Next".

Exception flow	UI overlays block ROI → auto-reposition; repeated misclassifications → suggest micro-tutorial.
Postconditions	Feedback delivered; learner state updated (accuracy, streaks, mastery counters).

Table 19: Use Case Descriptions 5

Field	**Description**
Use case name	Upload Learning Content
Description	Teacher adds or updates lesson items (letters/words/objects) with labels and age-appropriate notes.
Actors	Teacher, System
Preconditions	Teacher authenticated with authoring permission.
Trigger	Teacher selects "Add content" and uploads asset/metadata.
Main flow	Validate file & labels → save as draft → publish to catalog.
Exception flow	Invalid labels/corrupt media → show errors and corrective hints.
Postconditions	New content appears in the learner catalog for practice.

Table 20: Use Case Descriptions 6

Field	**Description**
Use case name	View Child Progress Report
Description	Parent views the child's progress summary with strengths and suggested practice areas.
Actors	Parent, System

Preconditions	Parent authenticated and linked to child account (consent verified).
Trigger	Parent opens “My Child’s Progress”.
Main flow	System renders progress cards, trends, and suggestions; parent reviews/downloads.
Exception flow	No recent activity → show guidance and encouragement tips.
Postconditions	Parent informed; optional PDF downloaded.

Table 21: Use Case Descriptions 7

4.10 Requirements

The functional requirements are prioritized using the “MoSCoW” method. This method considers critical elements such as the delivery schedule, potential risks, and the value these features bring to the stakeholders.

Priority Level	**Description**
M – Must Have	Non-negotiable requirements that are mandatory for the prototype.
S – Should Have	Important features that are not essential for the prototype but would significantly enhance its value.
C – Could Have	Features that are nice to have but have relatively less impact on the overall system.
W – Will Not Have	Features that are not planned to be included in the current system scope due to lower priority or minimal impact.

Table 22: MoSCoW Priority levels

4.10.1 Functional Requirements

ID	Requirement	Priority
FR01	System shall allow users (learner/teacher/parent) to register and log in using a role-based access.	Must
FR02	The system will show tutorial demos (video/animation) on letters, numbers and chosen words prior to the practice.	Must
FR03	The platform will facilitate capture of webcams and gestures capture to practice.	Must
FR04	The system must be able to identify the static alphabet gestures in real-time and provide a predicted class and confidence value.	Must
FR05	The system will display real time correction feedback (text/icon/overlay tips) upon completion of individual attempt.	Must
FR06	Each attempt (item, confidence, result, latency, timestamp) shall be logged by the system and learner progress updated.	Must
FR07	At the end of practice, the learner will be shown a summary of the session (attempts, highest accuracy, mastered items).	Must
FR08	Teachers shall upload or curate learning items (letters/words/objects) with labels and age-appropriate notes.	Should
FR09	The platform will be able to handle per-letter feedback and fingerspelling sequences (short words, 2-4 letters).	Should
FR10	Parents/guardians will get access to the progress report of the child (read-only) after consent linking.	Should
FR11	The system will offer minimum gamification (streaks/badges/points) that are based on rules of mastery.	Could

FR12	The teacher is also supposed to export a learner or a class report (CSV/ PDF).	Could
FR13	The platform must have the on-screen instructions on the camera permission, lighting and hand placement.	Must
FR14	In the case of a camera access failure (no recognition, tutorials only), the platform will be put into the mode of demos only.	Must
FR15	The system will be anonymous in respect of the telemetry used in analytics and model improvement.	Must

Table 23: Functional Requirements

4.10.2 Non-Functional Requirements

NFR ID	Description	Specification	Priority Level
NFR 1	The system must be very efficient in that real-time gesture recognition should be done with no apparent lag.	Performance	Must Have
NFR 2	The interface is to be user-friendly, child-friendly, and provide DHH learners with low reading levels so that they can use the interface.	Usability / Accessibility	Must Have
NFR 3	The platform must be scalable to accommodate several simultaneous learners, teachers and parents without loss of performance.	Scalability	Should Have

NFR 4	The system must protect user information with encrypted authentication, HTTPS hostility as well as parental-consent processes.	Security / Privacy	Must Have
NFR 5	It should be compatible with the popular web browsers (Chrome, Edge, Firefox, Safari) and must be able to scale to various screen sizes.	Compatibility	Should Have
NFR 6	This system must be available 99% of the time when in the state of operation and be able to withstand the loss of a connection gracefully.	Reliability / Availability	Should Have
NFR 7	The system should maintain data integrity and allow administrators to manage content without disrupting user activity.	Maintainability	Should Have
NFR 8	The platform is to be such that recognition performance can be reproduced between environments and devices.	Consistency	Should Have

Table 24: Non-Functional Requirements

4.11 Chapter Summary

This chapter described the software requirements specification of the proposed AI-based web-based system sign language learning. It outlined the process of collecting and analysing the needs of the stakeholders, which involved literature reviews, interviews, surveys, and brainstorming. The visual tool that was employed to conceptualize the system operational environment and determine the roles of the critical stakeholders like learners, teachers, and administrators was a rich picture diagram and stakeholder onion model. The concept of requirement elicitation assisted the definition of the functional and non-functional features of the system with the aid of the context and use case diagrams that demonstrated how the system was going to interact. Lastly, MoSCoW prioritization method was used to guarantee that the key features like real-time gesture recognition, feedback systems, and accessibility should be created initially, which would become a strong base of the system design and implementation processes in subsequent stages.

CHAPTER 05: TIME SCHEDULE

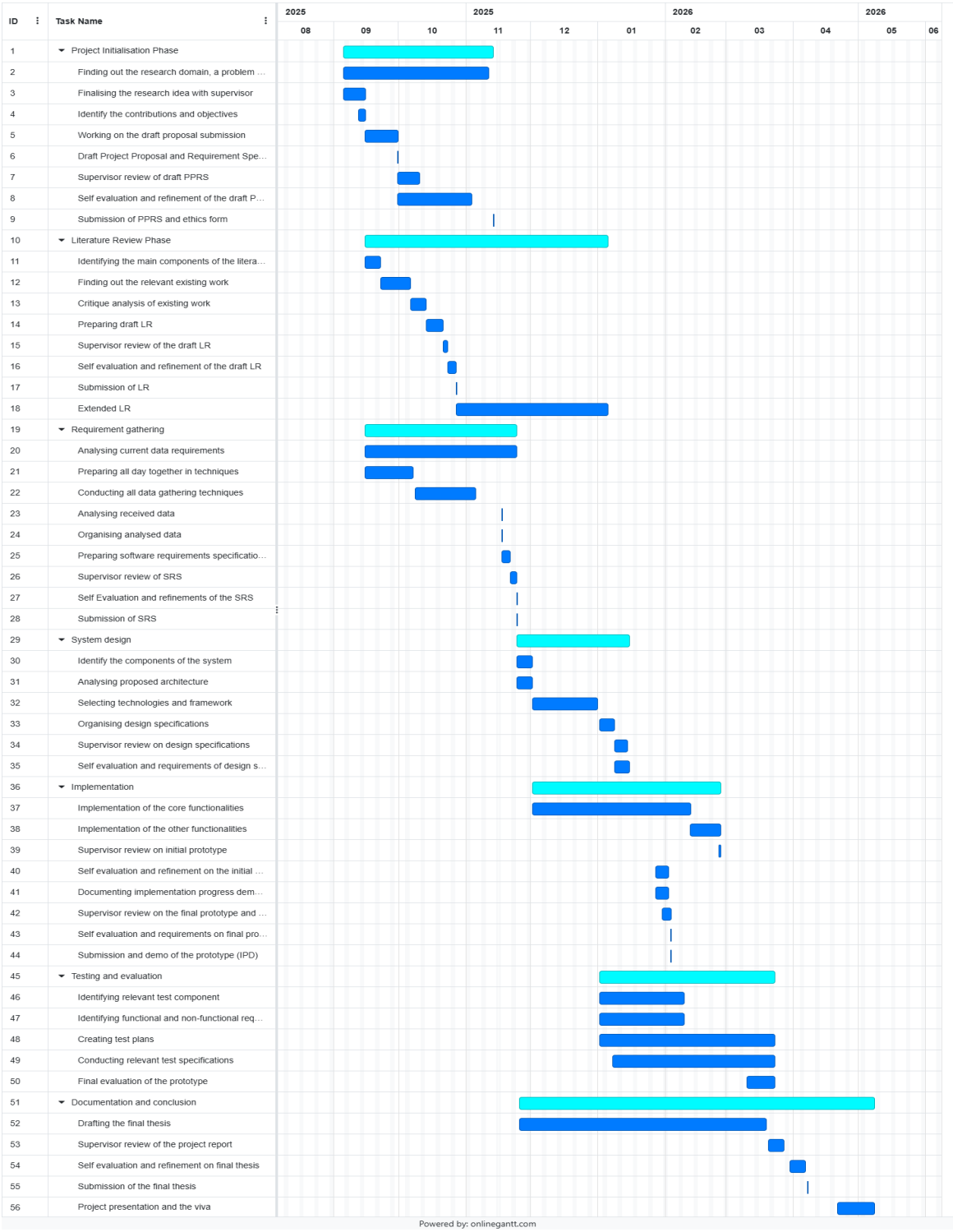

Figure 8: Complete Gantt Chart

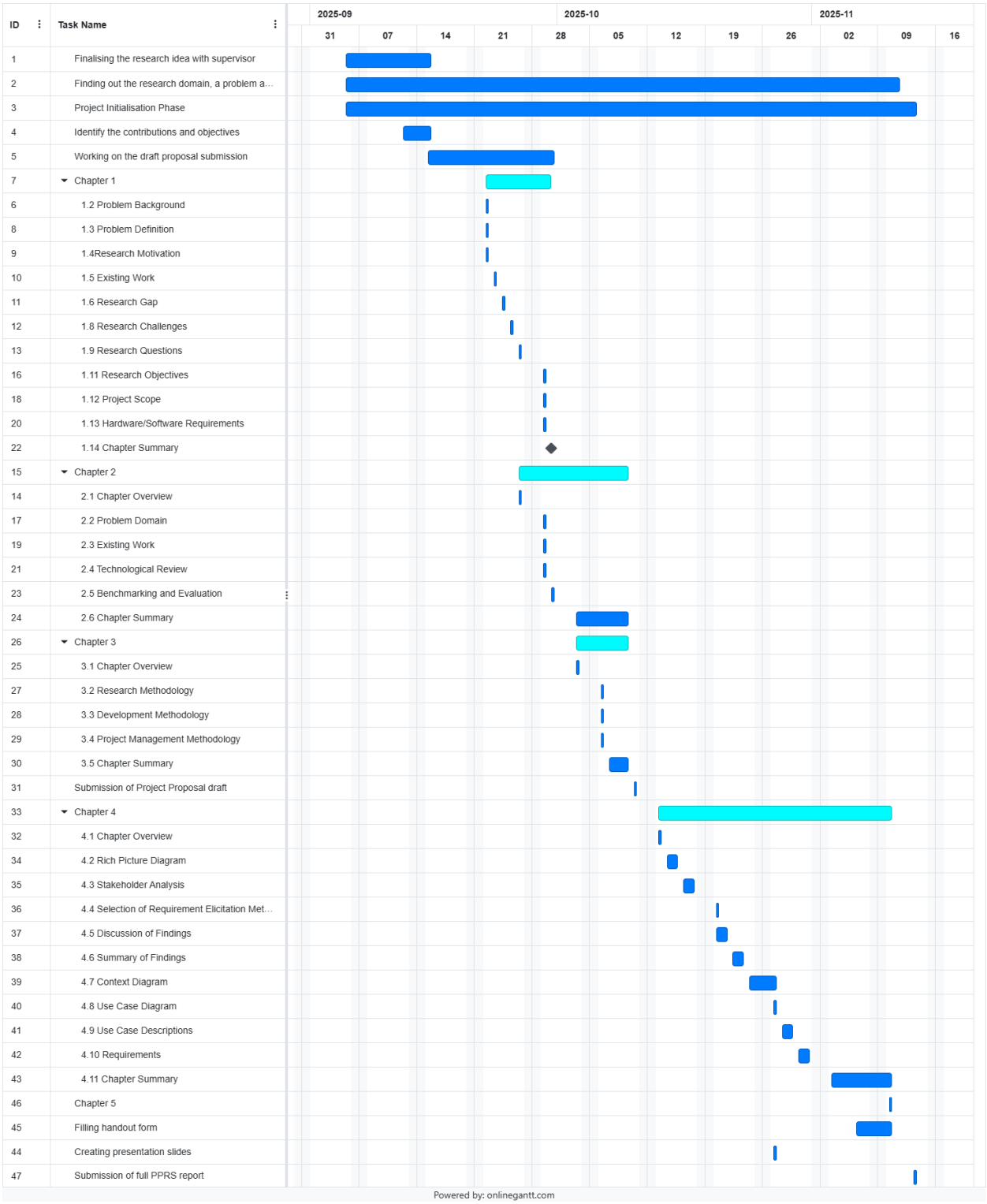

Figure 9: PPRS Timeline

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